

AI-Powered B2B Outreach & Lead Generation Platform

Product Requirement & System Workflow Document (PRD)

I. Product Overview

Objective

Build an AI-powered outreach and lead generation platform that automates:

- Company research
- Lead qualification
- Contact discovery
- Email verification
- Personalized outreach
- Follow-up automation
- Conversation tracking
- Meeting scheduling
- Customer notifications

The platform should operate as an end-to-end intelligent sales outreach assistant while still allowing user review and control.

2. User Inputs

The platform will collect the following inputs from customers:

Company Target Data

- CSV file containing target companies

Product Information

- Product/service details
- Value proposition
- Industry category
- Key differentiators

Target Customer Requirements

- Ideal customer profile
- Industry preference
- Geography preference (optional)
- Company size preference
- Business requirements
- Customer-defined ranking criteria

Outreach Configuration

- Email template/format (Optional)
 - Email footer/signature
 - Tone preference
 - Outreach personalization level
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3. Backend Workflow

Phase I: Company Intake & Research

Objectives

Process uploaded company data and gather intelligence.

Workflow

Company Processing

- Parse uploaded CSV
- Validate company data
- Remove duplicates
- Normalize company information

Automated Company Research

AI will collect:

- Company profile
 - Industry
 - Products/services
 - Business model
 - Company size
 - Public information
 - Website insights
 - Market relevance
 - Recent Funding
 - Recent Negative News
 - Active Hiring or not (Active Hiring is a positive sign)
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Phase 2: Company Scoring & Ranking

Objective

Rank companies based on customer requirements.

Weighted Scoring Parameters

Possible factors:

- Product fit
- Industry alignment
- Company relevance
- Requirement satisfaction
- Market compatibility
- Growth indicators

Workflow

Platform will:

1. Assign weighted score
2. Generate explainable scoring
3. Rank all companies
4. Select top N companies specified by customer

Phase 3: Contact Discovery

Objective

Identify relevant decision-makers.

Workflow

For each shortlisted company:

Contact Search

Find:

- Top 3 sales contacts
- Decision makers
- Relevant business personnel

Contact Information

Collect:

- Name
- Role
- Linked profile (if available)
- Company association

Phase 4: Email Discovery & Verification

Objective

Generate and verify email addresses.

Workflow

Email Guessing Logic

Generate emails using:

- Company domain

- Standard patterns
- Name-domain combinations

Examples:

- `firstname@company.com`
- `firstname.lastname@company.com`
- `f.lastname@company.com`

Email Verification

Use Verifalia.

Logic:

- Verify guessed emails
- Stop guessing once verified email found
- Mark verification status

Possible states:

- Verified
 - Risky
 - Invalid
 - Unknown
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Phase 5: AI Outreach Generation

Objective

Create personalized outreach. Use AI API key

Email Generation Inputs

- Customer template (optional)
- Customer product details
- Company research
- Contact role
- Email footer

Workflow

Platform will:

- Personalize outreach
 - Generate subject line
 - Generate body content
 - Maintain tone consistency
 - Create recipient-specific messaging
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Phase 6: Automated Outreach Execution

Objective

Send emails.

Workflow

- Use verified email only
- Send outreach
- Track delivery status
- Store thread history

Possible states:

- Queued
 - Sent
 - Delivered
 - Failed
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Phase 7: Email Monitoring & Follow-Up Automation

Objective

Automate conversation management.

Tracking Frequency

Every 15 minutes.

Workflow

System will:

- Monitor inbox
- Detect unread/unanswered emails
- Analyze prior thread
- Generate contextual follow-up
- Send intelligent replies

Continue until:

- Meeting scheduled
 - Campaign stopped
 - Lead closed
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Phase 8: Meeting Coordination

Objective

Manage meetings.

Workflow

When meeting fixed:

Platform will:

- Capture meeting link
 - Store meeting details
 - Notify customer
 - Notify company contact
 - Maintain meeting history
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4. Frontend Requirements

Module I: Authentication

Login Page

Features:

- Email login
- Password login
- Forgot password

Sign-Up Page

Features:

- Email registration
- Password creation
- OTP email verification
- Google Sign-Up (subject to free-tier support)

Security

- Session management
 - OTP validation
 - Authentication protection
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Module 2: Dashboard (Home Page)

Purpose

Primary control center.

Dashboard Widgets

Campaign Overview

- Active campaigns
- Completed campaigns
- Paused campaigns

Outreach Metrics

- Companies uploaded
- Companies researched

- Emails sent
- Replies received
- Meetings booked

Conversion Funnel

- Upload → Qualified → Contacted → Replied → Meeting

Activity Feed

Recent actions:

- Email sent
 - Reply received
 - Meeting fixed
 - Follow-up triggered
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Module 3: Campaign Management

Purpose

Manage outreach campaigns.

Features

Campaign Operations

- Create campaign
- View campaigns
- Pause campaign
- Resume campaign
- Duplicate campaign
- Archive campaign
- Delete campaign

Campaign Status

- Draft
- Running
- Paused
- Completed
- Failed

Module 4: New User / Campaign Creation Form

Upload Section

- Upload CSV
- Download sample CSV
- Validate format

Product Details

- Product description
- Value proposition
- Industry

Requirement Details

- Target customer requirements
- Ranking criteria

Outreach Settings

- Optional email template
- Footer
- Tone selection

Module 5: Company Research & Ranking Interface

Company List View

Display:

- Company name
- Industry
- AI score
- Rank

- Match level
- Status

Company Detail View

Display:

- Research summary
- Match explanation
- Scoring breakdown
- Contacts discovered
- Verification progress

Actions:

- Approve
- Exclude
- Re-rank

Module 6: Contact Discovery Review

Purpose

Human approval layer.

Display

For each company:

- Contact name
- Position
- Email
- Verification status
- Confidence score

Actions

- Approve
 - Reject
 - Edit contact
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Module 7: Email Draft Review & Editor

Purpose

Review AI-generated outreach.

Features

Email Preview

- Subject line
- Body
- Footer
- Personalization preview

Editor

- Rich text editing
- Manual edits
- AI regenerate

Testing

- Send test email
 - Recipient preview
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Module 8: Conversation / Inbox Interface

Purpose

CRM-style communication center.

Features

Display:

- Sent emails
- Replies
- Thread history
- Follow-up history
- AI reply suggestions

- Lead progress

Navigation:

Campaign → Company → Contact → Email Thread

Module 9: Meeting Management

Purpose

Meeting tracking.

Features

- Upcoming meetings
 - Meeting history
 - Meeting status
 - Calendar view
 - Meeting notes
 - Join link
 - Reminder system
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Module 10: Notifications Center

Purpose

Real-time updates.

Notification Types

- New reply
- Meeting scheduled
- Verification failed
- Campaign completed
- Follow-up sent
- System errors

UI:

- Bell icon
 - Notification panel
 - Mark as read
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Module 11: Agents Section

Purpose

AI workflow management.

Features

- Agent configuration
 - Workflow editing
 - Automation settings
 - Personalization controls
 - Edit details
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Module 12: Settings

Profile Settings

- Name
- Email
- Password change

Email Settings

- Sender email
- Signature defaults
- SMTP/API settings

AI Settings

- Tone
- Follow-up timing

- Personalization level

Security Settings

- OTP
 - Session control
 - Login history
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Module 13: Billing & Subscription (Future Ready)

Features

- Current plan
 - Usage limits
 - Email credits
 - Payment history
 - Upgrade plans
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Module 14: Integrations

Potential Integrations

- Email providers
 - Calendar systems
 - CRM export
 - Meeting tools
 - API/Webhooks
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Module 15: Activity Logs & Audit Trail

Purpose

Transparency and debugging.

Logs

- Campaign actions
 - Email events
 - AI actions
 - Verification attempts
 - User changes
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Module 16: Support Pages

About Us

- Product mission
- Team
- Features

Contact Us

- Support form
 - Email support
 - FAQ links
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Module 17: Error Handling & Empty States

Error Handling

Clear UI for:

- Invalid CSV
- Missing data
- Email failure
- Verification failure
- API timeout

Empty States

Provide:

- Guided onboarding
- Tooltips
- Demo campaign
- Sample workflows

Module 18: Admin Panel (Optional)

For platform administrators.

Features

- User management
- Campaign monitoring
- Support dashboard
- System logs
- Abuse detection

5. End-to-End Workflow Summary

User Journey

Sign Up → Create Campaign → Upload Companies → Research → AI Scoring → Contact Discovery → Email Verification → Email Review → Outreach → Conversation Tracking → Follow-Up Automation → Meeting Scheduling → Customer Notification

6. Suggested Navigation Structure

Authentication

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|— Dashboard

|— Campaigns

| |— Create Campaign

| |— Company Research

| |— Contacts

| |— Email Review

| |— Conversations

|— Meetings

1. Final Tech Stack

- Frontend stack
- Backend stack
- Database
- Authentication
- Email infrastructure
- AI layer
- Search + scraping stack
- WebSocket/realtime stack
- Hosting & deployment
- Free-tier services

2. System Architecture

- Full system flow
- Component interaction
- Request lifecycle
- Data flow

3. Multi-Agent Architecture

Separate agents for:

- Company Enrichment
- Company Scoring
- Employee Finder
- Email Guessing
- Email Verification
- Outreach Generation
- Email Tracking & Follow-up
- Meeting Coordination

including how they communicate sequentially.

4. Database Design Overview

Core schemas/tables:

- Users
- Campaigns
- Companies
- Contacts
- Email Threads
- Meetings
- Notifications
- Logs

5. API & Backend Design

- FastAPI services
- REST/WebSocket layer
- Background job handling
- Gmail + Calendar integration
- DuckDuckGo + scraping pipeline

6. Frontend Architecture

- Next.js structure
- Pages
- Dashboard flow
- Auth flow
- State management
- Component strategy

7. Deployment & DevOps

Free-tier deployment:

- Frontend
- Backend
- PostgreSQL
- Worker deployment
- Environment variables
- CI/CD suggestion

8. Suggested Folder Structure

For:

- Frontend
- Backend
- Agent modules
- Shared configs